

WUTAI SHAN, China

WMO: 535880

Lat: 38.9417N

Lon: 113.5133E

Elev: 2210

StdP: 77.44

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-20.9	-34.6	0.2	-16.2	-32.1	0.2	-15.4	23.7	-23.2	20.0	-22.5	10.5	270	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	21.2	14.1	20.0	13.7	18.9	13.2	17.2	18.9	16.2	18.0	15.3	17.2	4.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	15.6	18.0	15.6	14.6	17.2	14.6	13.7	16.4	58.4	19.0	54.8	18.1	51.5	17.2	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4)	17.1	14.4	12.0	-26.7	23.8	3.5	1.8	-29.2	25.1	-31.3	26.1	-33.3	27.1	-35.8	28.4	
(5)				-27.0	18.7	3.5	1.1	-29.5	19.5	-31.5	20.1	-33.5	20.8	-36.0	21.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-11.9	-9.3	-3.7	3.5	9.1	13.0	15.3	13.9	9.7	3.6	-4.2	-10.5	(6)
(7)		DBStd	10.45	4.97	4.99	5.33	5.14	3.83	2.85	2.34	2.55	3.10	4.11	4.90	4.78	(7)
(8)		HDD10.0	3229	679	542	424	202	63	8	0	3	42	202	427	636	(8)
(9)		HDD18.3	5818	938	775	682	447	287	162	98	139	261	458	677	894	(9)
(10)		CDD10.0	457	0	0	0	6	34	97	163	123	32	2	0	0	(10)
(11)		CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0	(11)
(12)		CDH23.3	6	0	0	0	0	0	2	4	0	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.5	6.4	6.3	6.3	6.1	5.5	4.5	3.9	3.9	4.1	5.3	6.4	6.8	(14)
(15)	Precipitation	PrecAvg	522	5	9	14	26	40	76	130	110	65	29	14	5	(15)
(16)		PrecMax	678	17	27	55	61	72	180	254	244	117	102	52	23	(16)
(17)		PrecMin	350	0	0	1	7	10	34	64	37	18	6	0	0	(17)
(18)		PrecStd	82	4	6	14	13	15	32	42	44	27	21	15	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	4.2	11.7	18.4	20.7	22.4	23.4	21.6	19.0	14.9	8.5	2.1	(19)
(20)			MCWB	-4.3	-1.8	2.8	6.9	9.2	11.7	16.1	15.5	11.1	7.5	1.1	-3.7	(20)
(21)		2%	DB	-1.5	1.3	8.4	15.2	18.4	20.3	21.5	20.1	17.1	12.6	5.9	-1.0	(21)
(22)			MCWB	-6.2	-3.6	0.7	5.5	8.5	11.7	15.2	15.0	10.8	6.4	-0.1	-5.5	(22)
(23)		5%	DB	-3.8	-0.8	6.0	13.0	16.7	19.0	20.4	19.0	15.5	10.9	4.0	-2.8	(23)
(24)			MCWB	-7.5	-4.6	-0.5	4.3	7.6	11.5	14.8	14.6	10.1	5.3	-0.8	-6.3	(24)
(25)		10%	DB	-5.6	-2.5	3.7	10.7	15.0	17.6	19.2	17.8	14.2	9.2	2.2	-4.4	(25)
(26)			MCWB	-8.7	-5.9	-1.6	3.3	6.8	10.9	14.5	14.2	9.6	4.4	-1.6	-7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.8	-1.0	3.7	9.5	12.2	15.4	18.9	18.4	14.5	9.7	2.9	-3.1	(27)
(28)			MCDWB	0.8	1.6	8.6	12.7	16.0	18.5	21.1	20.1	16.3	12.8	5.1	-0.1	(28)
(29)		2%	WB	-5.9	-2.8	1.5	7.3	10.6	14.2	17.7	17.1	13.0	7.8	1.2	-4.8	(29)
(30)			MCDWB	-2.4	0.2	7.3	12.9	15.1	17.4	19.4	18.5	14.5	10.9	4.0	-1.4	(30)
(31)		5%	WB	-7.2	-4.3	0.0	5.7	9.4	13.3	16.7	16.1	11.7	6.5	-0.2	-6.2	(31)
(32)			MCDWB	-4.0	-1.0	5.4	11.4	14.1	16.6	18.6	17.5	13.8	9.5	3.2	-3.2	(32)
(33)		10%	WB	-8.5	-5.8	-1.4	4.3	8.3	12.4	15.8	15.2	10.6	5.2	-1.4	-7.3	(33)
(34)			MCDWB	-5.8	-2.7	3.4	9.4	12.9	15.6	17.7	16.7	13.0	8.1	1.9	-4.6	(34)
(35)	Mean Daily Temperature Range	MDBR	6.5	6.8	7.6	8.2	8.0	7.0	5.9	5.8	6.0	6.5	6.5	6.3	(35)	
(36)		5% DB	MCDBR	7.4	8.0	9.3	10.2	9.6	8.8	7.5	7.0	7.2	7.9	7.5	7.0	(36)
(37)			MCWBR	5.3	5.6	5.5	5.4	5.4	5.1	5.2	4.9	4.9	4.7	4.9	5.0	(37)
(38)		5% WB	MCDBR	7.1	7.5	9.0	9.4	8.5	7.4	6.3	5.9	6.1	6.9	6.8	6.7	(38)
(39)			MCWBR	5.4	5.6	5.8	5.8	5.5	5.1	5.0	4.6	4.9	4.7	4.9	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.342	0.359	0.400	0.431	0.516	0.508	0.496	0.439	0.387	0.306	0.281	(40)	
(41)		taud	2.313	2.150	2.143	2.089	2.038	1.855	1.925	1.933	2.035	2.115	2.318	2.364	(41)	
(42)		Ebn at Noon	887	866	892	875	852	778	780	778	798	801	852	866	(42)	
(43)		Edh at Noon	91	124	139	157	169	203	188	182	154	127	90	81	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.69	3.40	4.70	5.51	6.11	5.43	5.01	4.75	4.15	3.50	2.75	2.40	(44)	
(45)		RadStd	0.22	0.26	0.39	0.37	0.49	0.44	0.30	0.39	0.33	0.30	0.35	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	-1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page